Course Project

Build your Interplanetary Mission Analysis Design tool.

- **Objective:** To compute the ΔV budget trading off different departure and arrival dates.

Design, implement and test a Pork Chop Plot tool by the building blocks:

- Planetary positions and velocities at a given departure and arrival dates.
- Solve the Lambert problem (any algorithm is OK) given the $\boldsymbol{r_1}$, $\boldsymbol{r_2}$ and specified Δt with the following steps:
  - Verify elemental cases against examples present in the GUI and/or in the theory slides
  - Implement a planar PCP ignoring the inclination and eccentricity of the departure/arrival planets
  - Implement a planar PCP assuming the actual planets eccentricities
  - Implement the actual PCP with actual inclinations and eccentricities
- Compute the required $\Delta V_T = |\Delta V_1| + |\Delta V_2|$ to embark the transfer orbital segment and to reach the target planet
- Generate the PCP and plot the transfer orbit

Deadline: *One week before the Final Exam*
Maximum Pages: *50*
Groups of three